

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4-ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario Based Questions	Assessment:	Assessment Tool 1– Scenario Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
Student Name:	T.Chandana	Reg. No.:	192472059
Section / Batch:	2024	Date:	17-08-2026

Code of Conduct

I T.Chandana (192472059)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___ T.Chandana _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Answer:

Given

- Claimed satisfaction = 80%
- Confidence level = 95%
- Confidence interval = 74% to 79%

Step 1: State the hypotheses

Null Hypothesis (H_0): At least 80% of students are satisfied.

$H_0: p \geq 0.80$

Alternative Hypothesis (H_1): Less than 80% of students are satisfied.

$H_1: p < 0.80$

Step 2: Compare the claim with the confidence interval

The 95% confidence interval is:

$74\% \leq p \leq 79\%$

The claimed value is: 80%

But 80% is outside the confidence interval.

Also, even the upper limit of the interval is only 79%, which is less than 80%.

Step 3: Decision

Since 80% is not contained in the 95% confidence interval, there is evidence against the claim that at least 80% of students are satisfied.

The university's claim is not supported.

The confidence interval indicates that the true satisfaction proportion is likely between 74% and 79%, which is entirely below the claimed 80%.

2. A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Answer:

Given

- Claimed mean = 30 minutes
- Confidence level = 99%
- Confidence interval = 31.5 to 35.2 minutes

Step 1: State the hypotheses

Null Hypothesis (H_0):

$H_0: \mu = 30$ minutes

Alternative Hypothesis (H_1):

$H_1: \mu \neq 30$ minutes

Step 2: Compare the claimed value with the confidence interval

Confidence interval:

$$31.5 \leq \mu \leq 35.2$$

Claimed value:

30 minutes

The value 30 is outside the confidence interval.

Step 3: Decision

Since 30 minutes is not included in the 99% confidence interval, the sample does not support the developer's claim.

In fact, the entire interval is above 30 minutes.

The claim is not supported.

The data suggests that the true average usage time is likely between 31.5 and 35.2 minutes, which is greater than the claimed 30 minutes.

3. A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Answer:

Given

- Claimed detection rate = 95%
- p-value = 0.02
- Significance level = 5% = 0.05

Step 1: State the hypotheses

Null Hypothesis (H_0):

$H_0: p = 0.95$

The fraud detection rate is 95%.

Alternative Hypothesis (H_1):

$H_1: p \neq 0.95$

The fraud detection rate is different from 95%.

Step 2: Compare p-value with significance level

We have:

p-value = 0.02

$\alpha = 0.05$

Compare:

$0.02 < 0.05$

Step 3: Decision Rule

- If p-value $< \alpha \rightarrow$ Reject H_0
- If p-value $\geq \alpha \rightarrow$ Fail to reject H_0

Here:

$0.02 < 0.05$

Therefore:

Reject H_0

The null hypothesis is rejected.

There is statistically significant evidence that the fraud detection rate is different from the claimed 95%.

Conclusion: The company's claim of a 95% detection rate is not supported by the data at the 5% significance level.

4. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Answer:

Given

- Existing conversion rate = 12%
- New conversion rate = 15%
- p-value = 0.03
- Significance level = 0.05

Step 1: State the hypotheses

Because we want to know whether conversion improves:

Null Hypothesis (H_0):

$H_0: p \leq 0.12$

The new algorithm does not improve the conversion rate.

Alternative Hypothesis (H_1):

$H_1: p > 0.12$

The new algorithm improves the conversion rate.

Step 2: Compare p-value and α

p-value = 0.03

$\alpha = 0.05$

Therefore:

$$0.03 < 0.05$$

Step 3: Decision

Since:

$$p\text{-value} < \alpha$$

we reject H_0 .

Step 4: Interpret the result

The observed conversion rate increased from:

$$12\% \rightarrow 15\%$$

and the p-value is less than 0.05.

Therefore, the improvement is statistically significant.

Yes, the new algorithm significantly improves the conversion rate.

There is sufficient statistical evidence at the 5% significance level to conclude that the new recommendation algorithm improves conversion compared with the existing 12% rate.

5. A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Answer:

Given

- $p\text{-value} = 0.08$
- $\text{Significance level} = 5\% = 0.05$

Step 1: State the hypotheses

Null Hypothesis (H_0):

The new system does not reduce the average waiting time.

Alternative Hypothesis (H_1):

The new system reduces the average waiting time.

Step 2: Compare p-value with α

$p\text{-value} = 0.08$

$\alpha = 0.05$

Compare:

$0.08 > 0.05$

Step 3: Decision Rule

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} > 0.05 \rightarrow \text{Fail to reject } H_0$

Here:

$0.08 > 0.05$

Therefore:

Fail to reject H_0

No, the hospital should not conclude that the new system reduces waiting time.

The $p\text{-value}$ of 0.08 is greater than the significance level of 0.05. Therefore, there is insufficient statistical evidence to conclude that the AI-based scheduling system significantly reduces average waiting time.